

B.Tech IV Year I Semester (R20) Regular &amp; Supplementary Examinations December/January 2024/2025

**GEOMETRIC DIMENSIONING AND TOLERANCES**

(Mechanical Engineering)

Time: 3 hours

Max. Marks: 70

**PART – A**

(Compulsory Question)

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- 1 Answer the following: (10 X 02 = 20 Marks)
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|----------------------------------------------------------|----|
| (a) Define limits.                                       | 2M |
| (b) What are the applications of simulators?             | 2M |
| (c) Give a brief note on principles of dimensioning.     | 2M |
| (d) List the steps involved in measuring the roundness.  | 2M |
| (e) Define runout.                                       | 2M |
| (f) What is location measurement?                        | 2M |
| (g) Differentiate between primary and secondary texture? | 2M |
| (h) What are the methods to measure Surface finish?      | 2M |
| (i) What is statistical tolerancing?                     | 2M |
| (j) List the applications of CMM.                        | 2M |

**PART – B**

(Answer all the questions: 05 X 10 = 50 Marks)

- 2 What is datum? Explain the role of datum systems. 10M
- OR**
- 3 Explain in detail the procedure involved in the inspection of dimensional and geometrical deviations. 10M
- 4 What is form tolerance? Explain with suitable examples. 10M
- OR**
- 5 Explain various types of orientation tolerances. Also explain their specifications and interpretations. 10M
- 6 Explain various types of profile tolerances. Also explain their specifications and interpretations. 10M
- OR**
- 7 Discuss in detail the verification techniques. Give suitable examples. 10M
- 8 How is surface roughness calculated by CLA and R.M.S methods? 10M
- OR**
- 9 The heights of peaks and valleys of 20 successive points on a surface are 35, 25, 40, 22, 37, 19, 41, 21, 42, 18, 42, 24, 44, 25, 40, 18, 40, 18, 39, and 21 microns respectively, measured over a length 20 mm. Determine CLA and RMS values of roughness surface. 10M
- 10 Write a detailed note on computer aided tolerancing and verification. 10M
- OR**
- 11 Explain the following: 10M
- (i) Working principle of CMM.
- (ii) Dimensional Chains.

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